

Opposition-Aware Graph Gating and Local Region Pooling for Object-Agnostic Zero-Shot State Recognition

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Abstract—Object state recognition aims to identify conditions such as *open*, *closed*, *empty*, or *filled*, but this becomes challenging when the object or its state has not been observed during training. While earlier approaches relied on known object categories or state–object compositions, recent object-agnostic zero-shot methods such as OaSC use knowledge graphs to recognize unseen states without explicitly identifying the object. Building on this idea, we propose *Opposition-Aware Contrastive Graph Gating (OCGG)* to refine noisy semantic neighborhoods by favoring concepts that better distinguish a state from its semantic opposite, together with *Local Region Pooling (LRP)* to capture informative spatial evidence that may be weakened by global feature pooling. The proposed approach retains the original Tr-GCN and ResNet-101 framework while improving its semantic and visual reasoning. Experiments on OSDD and MIT-States show that OCGG with LRP improves the harmonic mean from 50.07% to 51.51% and from 51.37% to 57.23%, respectively, demonstrating the benefit of combining contrastive semantic knowledge with localized visual cues for object-agnostic zero-shot state recognition.

Index Terms—object state recognition object-agnostic recognition, knowledge graphs, local region pooling, visual-semantic learning

I. INTRODUCTION

Recognizing an object is only part of understanding a scene. In many applications, the state of the object determines what can happen next: a bottle may be *open* or *closed*, a container *empty* or *filled*, and a piece of fabric *folded* or *unfolded*. Such state information is important for reasoning about affordances, interactions, task progress, and possible future actions [1], [2]. This makes object-state recognition particularly relevant to robotics, assistive systems, scene understanding, and other applications in which an agent must reason not only about what an object is, but also about its current condition.

Object-state recognition is challenging because state appearance is often less stable than object appearance. The same state can look very different across object categories, while two states of the same object may differ only in a small visual detail [1], [3]. An open bottle, an open drawer, and an open book, for example, share the same semantic state while having very different appearances. The Object State Detection Dataset (OSDD) study showed that state detection is generally harder than conventional object detection, despite involving fewer

state classes, highlighting the need for methods that learn state-specific rather than object-specific cues [1].

Previous work has gradually relaxed the assumptions under which states are recognized. Jelodar and Sun [4] jointly predicted objects and their states and showed that language knowledge from ConceptNet could improve visual predictions. However, their method operates in a supervised closed-set setting in which both object and state vocabularies are known during training. Compositional zero-shot learning (CZSL) later addressed unseen combinations of known state and object primitives [5]. Its open-world extension further removed the assumption that the set of possible state–object compositions is known at test time [6]. Nevertheless, these approaches still treat the object as part of the prediction and assume that the individual state and object primitives have been observed during training.

A more recent formulation, Object-agnostic State Classification (OaSC), removes the need to identify the object altogether. Goudis *et al.* [2] use common-sense knowledge graphs together with a Transformer Graph Convolutional Network (Tr-GCN) to generate classifier weights for both seen and unseen states. The Tr-GCN architecture was originally introduced for zero-shot learning with common-sense knowledge graphs by Nayak and Bach [7]. These generated weights form the final classification layer of a ResNet-101 visual model [8], whose remaining layers are fine-tuned using images from seen states only [2]. The resulting classifier can therefore predict a state without explicitly using the object’s category and can operate in a generalized zero-shot setting containing both seen and unseen state classes.

Although knowledge graphs provide useful external information for zero-shot recognition [7], their semantic neighborhoods are not always ideal for visual state discrimination. General-purpose resources such as ConceptNet [9] and WordNet [10] encode broad commonsense and lexical relationships rather than being designed specifically for visual state discrimination. As a result, a concept can be strongly related to a state without helping to distinguish that state from its opposite. For example, *open* and *closed* are semantically related precisely because they describe opposite conditions. The ablation analysis in OaSC shows that adding excessive or irrelevant graph nodes can reduce performance and that

selective node inclusion is beneficial [2]. Similar behavior was observed in earlier ConceptNet-based state recognition, where strong semantic associations could occasionally push an otherwise correct visual prediction toward an implausible state [4]. These observations motivate a more discriminative treatment of semantic neighborhood information.

A second limitation concerns spatial evidence. The standard OaSC visual classifier ultimately relies on a globally pooled representation of the object [2]. Although global pooling provides a compact representation for classification, convolutional feature maps can retain spatially localized class-discriminative information [11]. This distinction is particularly relevant to object states because useful evidence may be concentrated in only a small region, such as the opening of a container, a plug connection, or the folded edge of a material. Global pooling can weaken such localized evidence by averaging it together with less informative regions.

To address these two issues, we extend the OaSC framework in both its semantic and visual components. First, we introduce *Opposition-Aware Contrastive Graph Gating (OCGG)*. Instead of ranking a graph concept only by its semantic similarity to a target state, OCGG also considers its similarity to the corresponding opposite state. Using CLIP text embeddings [12], a graph neighbor is favored when it is more strongly associated with the target state than with its semantic opposite. This converts general semantic relatedness into a more discriminative signal before the graph is processed by the original Tr-GCN. Second, we introduce *Local Region Pooling (LRP)*, which retains the spatial feature map produced by ResNet-101 and evaluates state evidence over local regions. Motivated by the observation that convolutional feature maps retain localized discriminative information [11], the strongest local regions are aggregated and used as a residual correction to the original global prediction. This allows localized state evidence to contribute without replacing the baseline classifier or retraining the visual backbone.

We evaluate the proposed approach on OSDD [1] and MIT-States [3] under the generalized zero-shot setting, where both seen and unseen classes are included in the test-time label space [13], [14]. The model is trained using seen states and evaluated over both seen and unseen states without using object labels. Performance is measured using seen accuracy, unseen accuracy, their harmonic mean (HM), and the area under the seen–unseen trade-off curve (AUC), following standard generalized zero-shot evaluation principles [13], [14]. On OSDD, combining OCGG with LRP improves the harmonic mean from 50.07% for the reproduced OaSC baseline to 51.51%. On MIT-States, the same final configuration improves the harmonic mean from 51.37% to 57.23%, while also increasing the maximum unseen-state accuracy and AUC. These results indicate that opposition-aware semantic reasoning and localized visual evidence provide complementary improvements to graph-guided object-agnostic state recognition.

The main contributions of this work are:

- We analyze the semantic neighborhoods used by OaSC and introduce *Opposition-Aware Contrastive Graph Gat-*

ing (OCGG), which uses CLIP-based contrastive similarity between a target state and its semantic opposite to produce more discriminative graph neighborhoods.

- We introduce *Local Region Pooling (LRP)*, a lightweight spatial refinement that extracts state-specific evidence from the final convolutional feature map and combines it with the original global classifier score without retraining the backbone.
- We evaluate the semantic and spatial components through ablation studies on OSDD and MIT-States and show that their combination improves the reproduced OaSC baseline on both datasets, with the largest improvement observed on MIT-States.

The remainder of this paper reviews the relevant literature, describes the proposed semantic and spatial refinements, presents the experimental evaluation and ablation studies, and concludes with the main findings and limitations of the work.

II. RELATED WORK

Our work lies at the intersection of object-state recognition, compositional zero-shot learning, and knowledge-guided zero-shot classification. We review these areas according to how they treat object identity, unseen states, and external semantic knowledge.

A. Object-State Recognition and Datasets

Isola *et al.* [3] were among the first to study object states as a distinct visual recognition problem. They observed that variations within an object category, which conventional recognition systems often try to ignore, can encode meaningful physical transformations such as *sliced*, *melted*, or *ripe*. Their MIT-States dataset was constructed from 115 adjectives and 249 nouns, producing 2,550 selected adjective–noun queries. Images were collected from the Web and cleaned through crowdsourcing. The resulting setting tests whether state concepts learned from one set of nouns can transfer to previously unseen object categories.

Their experiments showed that state information can transfer across nouns, but that generalization becomes more difficult as visually or semantically similar training objects are removed. This established an important property of state recognition: a useful state representation should capture what is shared by a state across different object categories rather than simply memorizing object appearance.

Goudis *et al.* [1] later introduced the Object State Detection Dataset (OSDD), which provides 13,744 images and 19,018 annotations covering 18 household object categories and 9 state classes. The states include *open/closed*, *empty/containing liquid*, *empty/containing solid*, *plugged/unplugged*, and *folded/unfolded*. Unlike Web-collected state datasets, OSDD is derived from video frames and contains cluttered scenes, varying viewpoints, and objects undergoing manipulation.

Their experiments showed that state detection is more difficult than conventional object detection and that state classifiers can become strongly dependent on object appearance. Generalization to a new object category was particularly weak

when the new object differed substantially in appearance from those used during training [1]. These findings motivate state representations that are less dependent on object identity.

B. Joint and Compositional Object–State Recognition

Jelodar and Sun [4] explored joint object and state recognition in a supervised closed-set setting. Their approach uses a ResNet-50 with separate object and state classification heads and incorporates semantic relationships obtained from ConceptNet and Google N-grams. Language-based object–state probabilities are combined with CNN predictions using additional classifiers and a refinement mechanism.

The inclusion of language knowledge improved both state and object recognition, demonstrating that semantic relationships can complement visual evidence. However, the authors also observed that ConceptNet may contain misleading associations; semantically related concepts are not always visually plausible state descriptions. Their method also assumes that both object and state classes are known during training and inference [4]. This work therefore highlights both the usefulness and the potential noise of general-purpose semantic knowledge for state recognition.

Compositional Zero-Shot Learning (CZSL) relaxes the closed-set assumption by requiring models to recognize unseen combinations of known states and objects. For example, a model may learn *wet dog* and *ripe tomato* and later be asked to recognize *wet tomato*. Mancini *et al.* [6] showed that conventional CZSL evaluation still assumes prior knowledge of the set of candidate compositions. They introduced Open-World CZSL (OW-CZSL), where the prediction space contains all possible state–object combinations, including implausible compositions.

Their CompCos model embeds visual features and state–object compositions in a shared space and estimates the feasibility of unseen compositions. Feasibility is used either to suppress implausible predictions or to modify the training objective. The open-world evaluation revealed a substantial drop in existing CZSL methods; on MIT-States, for example, the unseen accuracy of LabelEmbed+ fell from 20.1% to 2.5%, while CompCos improved the open-world AUC over the previous best method from 0.8 to 1.6 [6].

Despite these advances, CZSL remains object-aware: unseen *combinations* are recognized from state and object primitives that have individually been observed during training. The object category also remains part of the predicted composition.

C. Object-Agnostic Zero-Shot State Recognition

Goudis *et al.* [2] introduced Zero-shot Object-agnostic State Classification (OaSC), in which the goal is to predict the state of an object without using or estimating its object category. In addition, some state classes evaluated at test time are not visually observed during training.

OaSC constructs a common-sense knowledge graph from resources such as ConceptNet and WordNet and uses a Transformer Graph Convolutional Network (Tr-GCN), based on the zero-shot graph framework of Nayak and Bach [7], to generate

classifier parameters for both seen and unseen states. These generated parameters form the final classification layer of a ResNet-101. The remaining visual layers are then fine-tuned using only images belonging to seen states while the generated classification layer is kept fixed. At inference time, the same classifier predicts over both seen and unseen state classes [2].

The method was evaluated on OSDD, MIT-States, and state-specific subsets of C-GQA and VAW and was shown to outperform several object-aware CZSL approaches when object information was removed. Its ablation studies are particularly relevant to our work. ConceptNet generally provides stronger state representations than WordNet, while combining the two sources can provide additional gains. More importantly, increasing the number of graph hops or indiscriminately adding neighboring nodes can degrade performance. Selective inclusion of relevant nodes improves results, indicating that semantic graph noise directly affects the quality of generated zero-shot classifiers [2].

D. Language-Guided Semantic Representations

Large-scale vision–language models provide another source of semantic knowledge. CLIP [12] learns a shared representation between images and natural-language descriptions from large-scale image–text supervision. Its text encoder provides embeddings in which semantically related concepts can be compared directly using cosine similarity.

In our setting, CLIP is not used as the final image classifier. Instead, we use its text representation as an additional semantic signal for evaluating graph neighbors. This is useful because graph connectivity represents broad lexical or commonsense relatedness, whereas a language embedding can provide a complementary measure of how strongly a neighboring concept corresponds to a target state. However, simple similarity to a state is still insufficient when closely related concepts represent opposite conditions, such as *open* and *closed*. This motivates explicitly contrastive semantic reasoning between opposing states.

E. Positioning of Our Work

The literature exposes two limitations that directly motivate our method. First, common-sense knowledge graphs contain useful relationships but also include concepts that are semantically related without being discriminative for visual state recognition. This problem is especially important for opposite states, whose names and surrounding concepts are often highly related despite representing different visual conditions. OaSC itself shows that filtering graph neighborhoods can improve zero-shot performance [2], while earlier ConceptNet-based state recognition similarly reports misleading semantic associations [4].

Second, state evidence may be spatially localized. OaSC ultimately classifies globally pooled ResNet features, although cues such as an opening, a plug connection, a fold, or a visible content boundary may occupy only a small portion of an object. Aggregating the entire spatial feature map can therefore weaken the evidence most relevant to the state.

Our method addresses these two issues while retaining the graph-guided zero-shot formulation of OaSC. We introduce *Opposition-Aware Contrastive Graph Gating (OCGG)*, which evaluates graph concepts according to whether they support a target state more strongly than its semantic opposite, and *Local Region Pooling (LRP)*, which supplements the global state score with evidence from the most informative local regions of the convolutional feature map. The following section describes these components in detail.

III. PROPOSED METHOD

Our method extends the object-agnostic zero-shot state classification framework of OaSC [2] along two complementary directions. First, we refine the semantic neighborhoods used by the graph model so that state classifiers are generated from concepts that are more discriminative for the target state. Second, we recover localized state evidence from the spatial feature map of the visual backbone instead of relying only on global average pooled features. The final framework combines *Opposition-Aware Contrastive Graph Gating (OCGG)* with *Local Region Pooling (LRP)* while retaining the pretrained OaSC Tr-GCN and ResNet-101 components.

A. Problem Formulation

Let $\mathcal{S} = \mathcal{S}^S \cup \mathcal{S}^U$ denote the set of state classes, where $\mathcal{S}^S$ contains states observed during visual training and $\mathcal{S}^U$ contains unseen states for which no training images are available. Given a cropped image x containing a single object, the goal is to predict its state $s \in \mathcal{S}$ without using or estimating the object’s category.

During training, visual examples are available only for states in $\mathcal{S}^S$. At test time, both seen and unseen states are possible, corresponding to the generalized zero-shot learning (GZSL) setting. The prediction function is therefore

$$f(x) : \mathcal{X} \rightarrow \mathcal{S}^S \cup \mathcal{S}^U, \quad (1)$$

where no object-class label is provided to the model.

The principal difficulty is that classifiers for states in $\mathcal{S}^U$ must be constructed without visual examples of those states. Following OaSC, external semantic knowledge is used to generate state classifiers. Our contribution is to improve both the semantic information supplied to this classifier-generation process and the visual evidence used at inference.

B. OaSC Baseline and Framework Overview

OaSC [2] represents each state using a classifier vector generated from a common-sense knowledge graph. A Tr-GCN, following the knowledge-graph zero-shot formulation of Nayak and Bach [7], maps semantic graph information to the visual classifier space. The released OaSC state representation has 2049 dimensions, consisting of a 2048-dimensional classifier weight vector and one bias term. We write this representation for state s as

$$\theta_s = [\mathbf{w}_s; b_s], \quad \mathbf{w}_s \in \mathbb{R}^{2048}. \quad (2)$$

A ResNet-101 produces a global visual representation

$$\mathbf{f}(x) \in \mathbb{R}^{2048}. \quad (3)$$

The corresponding baseline state score is

$$z_s^{\text{global}}(x) = \mathbf{f}(x)^\top \mathbf{w}_s + b_s. \quad (4)$$

Equivalently, by appending a constant one to the visual representation, the score is the inner product between the augmented image feature and the 2049-dimensional state vector.

For our experiments, we retain the released graph structure, node features, Tr-GCN parameters, and ResNet-101 visual backbone. To enable controlled modification of graph neighborhoods, we reproduce the released Tr-GCN forward pass in PyTorch using its original checkpoint. The first graph stage maps the original 300-dimensional graph features to 2048 dimensions and samples the top 100 neighbors, while the second stage produces the 2049-dimensional classifier vectors using the top 50 neighbors.

Our proposed framework modifies only two parts of this pipeline. On the semantic side, the ranking of neighbors associated with each target state is refined before the unchanged Tr-GCN generates θ_s . On the visual side, the final convolutional feature map is retained and used by LRP to complement the global prediction in Eq. (4). Neither component requires retraining the Tr-GCN or the visual backbone.

Figure 1 summarizes the complete pipeline. The semantic and visual refinements are described next.

C. CLIP-Guided Semantic Reranking

The OaSC graph associates each state node s with a neighborhood $\mathcal{N}_s$ and an original random-walk relevance score $r_{s,n}$ for every neighboring concept n . These scores encode graph connectivity, but a highly connected concept is not necessarily useful for visually distinguishing the state.

We therefore introduce an additional semantic relevance signal using the text encoder of CLIP [12]. Importantly, CLIP is not used as the image classifier or visual backbone in our method. It is used only to embed state and graph-node descriptions.

For every graph concept n , we use the prompt

a visual concept related to <node>

and for each state s ,

a photo of an object that is <state>

The CLIP text embeddings are L_2 -normalized. Let $\mathbf{e}_n$ and $\mathbf{e}_s$ denote the embeddings of a neighbor and target state, respectively. Their semantic similarity is

$$c_{s,n} = \cos(\mathbf{e}_s, \mathbf{e}_n). \quad (5)$$

Before reranking, repeated occurrences of the same graph neighbor are deduplicated, retaining the occurrence with the largest original random-walk score. The graph relevance scores are then normalized within each state neighborhood.

As an intermediate baseline, we consider a fixed semantic fusion,

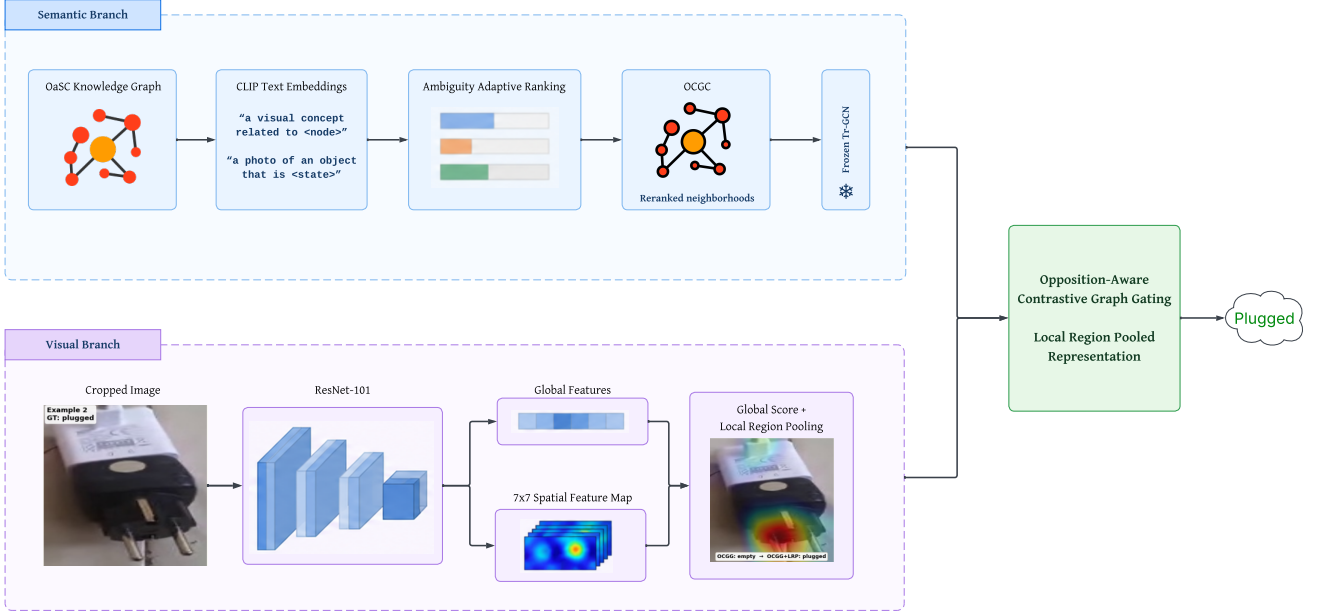


Fig. 1. Overview of the proposed framework. The semantic branch refines OaSC knowledge-graph neighborhoods using CLIP-based ambiguity-adaptive weighting and OCGG before generating state classifier weights with a frozen Tr-GCN. In parallel, the visual branch extracts global and spatial features with ResNet-101, where Local Region Pooling (LRP) captures localized state evidence. The global and LRP-based scores are then combined and calibrated for generalized zero-shot state prediction.

$$q_{s,n} = 0.2 \hat{r}_{s,n} + 0.8 \hat{c}_{s,n}, \quad (6)$$

where $\hat{r}_{s,n}$ and $\hat{c}_{s,n}$ denote normalized graph and CLIP relevance, respectively. This tests whether semantic reranking alone can improve the neighborhoods used by the Tr-GCN.

1) *Ambiguity-Adaptive Semantic Weighting*: A fixed semantic weight assumes that every state neighborhood is equally ambiguous. In practice, some states have only a few strongly ranked graph neighbors, whereas others contain many alternatives with similar random-walk scores. We therefore make the CLIP contribution state-dependent.

For state s with n_s unique neighbors, the original graph scores are converted to a distribution

$$p_{s,n} = \frac{\max(r_{s,n}, \epsilon)}{\sum_{j \in \mathcal{N}_s} \max(r_{s,j}, \epsilon)}. \quad (7)$$

We measure the normalized neighborhood entropy as

$$H_s = -\frac{\sum_{n \in \mathcal{N}_s} p_{s,n} \log p_{s,n}}{\log n_s}, \quad (8)$$

and combine it with a neighborhood-size factor

$$D_s = \frac{\log(1 + n_s)}{\log(1 + N_{\max})}, \quad (9)$$

where $N_{\max}$ is the largest state-neighborhood size. The resulting ambiguity score is

$$A_s = H_s D_s. \quad (10)$$

The semantic fusion weight is then

$$\lambda_s = \lambda_{\min} + A_s (\lambda_{\max} - \lambda_{\min}), \quad (11)$$

with $\lambda_{\min} = 0.40$ and $\lambda_{\max} = 0.90$. A larger, high-entropy neighborhood therefore receives stronger CLIP guidance, while a small or already decisive neighborhood remains more strongly influenced by its original graph relevance.

The adaptive reranking score becomes

$$q_{s,n}^{\text{adapt}} = (1 - \lambda_s) \hat{r}_{s,n} + \lambda_s \hat{c}_{s,n}. \quad (12)$$

We do not hard-prune the neighborhood after this operation. All unique neighbors are retained and reordered according to the new score. Consequently, the original Tr-GCN sampler still performs its top-100 and top-50 selections, but now from a semantically refined ranking.

D. Opposition-Aware Contrastive Graph Gating

Absolute semantic similarity does not necessarily provide a discriminative signal for state recognition. Antonyms such as *open* and *closed* are semantically related and may therefore both receive high similarity to the same graph concept. For state classification, however, the useful question is not simply whether a concept is related to *open*, but whether it supports *open* more strongly than *closed*.

We address this with *Opposition-Aware Contrastive Graph Gating* (OCGG). For states with an identified semantic opposite $\bar{s}$, we replace the absolute CLIP similarity with the contrastive margin

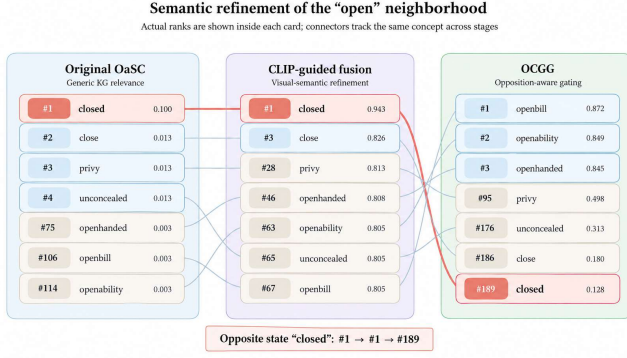


Fig. 2. Semantic refinement of the *open* neighborhood across three stages: the original OaSC ranking, CLIP-guided fusion, and the proposed OCGG reranking. The figure tracks how candidate neighbors evolve as visual-semantic evidence is incorporated, and shows that the opposite state *closed*, initially ranked highly, is strongly demoted by OCGG. This illustrates that OCGG favors concepts supporting the target state over those associated with its semantic opposite.

$$m_{s,n} = \cos(\mathbf{e}_s, \mathbf{e}_n) - \cos(\mathbf{e}_{\bar{s}}, \mathbf{e}_n). \quad (13)$$

A positive value indicates that concept n supports the target state more strongly than its opposite, while a negative value indicates that the concept is more characteristic of the opposite state.

The opposite-state pairs used by our method are

$$\begin{aligned} open &\leftrightarrow closed, & empty &\leftrightarrow filled, \\ folded &\leftrightarrow unfolded, & plugged &\leftrightarrow unplugged. \end{aligned} \quad (14)$$

An opposite state is used only as a semantic anchor and does not need to be part of the final prediction space. For a state without a natural opposite, such as *containing*, we retain the absolute CLIP similarity.

The contrastive values are normalized within each neighborhood and fused with the original graph relevance using the same ambiguity-adaptive weight from Eq. (11):

$$q_{s,n}^{\text{OCGG}} = (1 - \lambda_s) \hat{r}_{s,n} + \lambda_s \hat{m}_{s,n}. \quad (15)$$

The resulting score replaces only the ranking value associated with the target state’s neighborhood. The graph node features, all non-state neighborhoods, and the learned Tr-GCN parameters remain unchanged. Thus, OCGG acts as a soft semantic gate: it changes which neighbors are emphasized by the existing Tr-GCN rather than learning a new graph network.

After reranking, the unchanged Tr-GCN produces a new classifier vector

$$\boldsymbol{\theta}_s^{\text{OCGG}} = [\mathbf{w}_s^{\text{OCGG}}, b_s^{\text{OCGG}}], \quad (16)$$

which is used by the same ResNet feature extractor.

E. Local Region Pooling

OCGG changes the semantic state classifier, but the baseline visual score in Eq. (4) is still computed from a globally pooled representation. We therefore introduce *Local Region Pooling* (LRP) to recover spatially concentrated evidence from the last convolutional stage of ResNet-101.

Let

$$\mathbf{F}(x) \in \mathbb{R}^{C \times H \times W} \quad (17)$$

denote the final spatial feature map before global average pooling. In our implementation, $C = 2048$ and $H = W = 7$. Global average pooling gives

$$\mathbf{f}(x) = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W \mathbf{F}_{:,h,w}(x). \quad (18)$$

For every candidate state s , its classifier weight $\mathbf{w}_s$ is directly applied at each spatial location to produce a state-specific evidence map

$$M_s(h, w) = \mathbf{w}_s^\top \mathbf{F}_{:,h,w}(x). \quad (19)$$

The classifier bias is not included in this spatial map because it is already present in the global score and does not contain location information.

Rather than selecting isolated high-response pixels, we search for coherent local regions. A 3×3 average-pooling window with stride one is applied to M_s , producing a 5×5 region-score map. Let $\mathcal{T}_s$ denote the three highest-scoring regions. The local score is

$$L_s = \frac{1}{K} \sum_{r \in \mathcal{T}_s} R_s(r), \quad K = 3, \quad (20)$$

where $R_s(r)$ is the average evidence within region r .

Simply adding L_s to the baseline score would count information already represented by global average pooling. We therefore compute the visual evidence already contained in the global feature,

$$G_s^{\text{vis}} = \frac{1}{HW} \sum_{h,w} M_s(h, w), \quad (21)$$

and define the localization residual

$$\Delta_s = L_s - G_s^{\text{vis}}. \quad (22)$$

The final LRP-enhanced logit is

$$z_s^{\text{LRP}} = z_s^{\text{global}} + \alpha \Delta_s. \quad (23)$$

Thus, LRP does not replace the global OaSC prediction. It contributes only the additional evidence obtained by concentrating on the most informative local regions. The operation is applied independently to every state class and introduces no learned attention module or additional trainable parameters.

Our final configuration uses 3×3 regions, the top three regions, and $\alpha = 0.20$. The sensitivity of α and the transfer

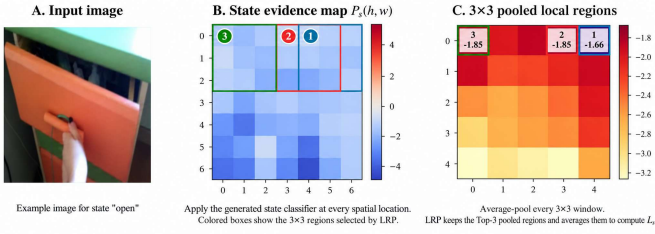


Fig. 3. Illustration of Local Region Pooling (LRP) for the state *open*. Given an input image (A), the generated state classifier is applied at every spatial location of the 7×7 ResNet feature map to obtain the state evidence map $M_s(h, w)$ (B). A 3×3 sliding average produces the 5×5 local-region score map (C), from which the top-3 regions are selected and averaged to obtain the local score L_s .

of this fixed configuration across datasets are examined in Section IV.

F. Generalized Zero-Shot Inference

Both the OaSC baseline and our proposed variants are evaluated in the generalized zero-shot setting, where seen and unseen states compete in the same prediction space. Zero-shot classifiers are commonly biased toward classes observed during visual training. Following the OaSC evaluation protocol [2], we therefore apply calibrated stacking to the unseen-state logits.

Let $z_s(x)$ denote the score produced by the selected model, which is Eq. (4) for the baseline and Eq. (23) for the final OCGG+LRP model. The calibrated score is

$$\tilde{z}_s(x; \gamma) = z_s(x) + \gamma \mathbb{I}[s \in \mathcal{S}^U], \quad (24)$$

where γ controls the trade-off between seen and unseen classes. A positive γ favors unseen states, while a negative value favors seen states. The final prediction is

$$\hat{s} = \arg \max_{s \in \mathcal{S}} \tilde{z}_s(x; \gamma). \quad (25)$$

For each operating point, we compute the seen-state accuracy $A_S(\gamma)$ and unseen-state accuracy $A_U(\gamma)$. Their harmonic mean is

$$HM(\gamma) = \frac{2A_S(\gamma)A_U(\gamma)}{A_S(\gamma) + A_U(\gamma)}. \quad (26)$$

Sweeping γ also produces the seen–unseen trade-off curve from which the area under the curve (AUC) is computed. The exact evaluation range, dataset splits, hyperparameter protocol, and ablation variants are described in the following section.

IV. EXPERIMENTAL SETUP

A. Datasets and Seen/Unseen Splits

We evaluate on two benchmarks used by the OaSC baseline, both adapted to the object-agnostic zero-shot state-classification protocol in which the object label is discarded and only the state label is predicted.

a) *OSDD*.: The Object State Detection Dataset [1] is a dataset explicitly designed for object-state detection. We use the cropped-object splits released with OaSC, which cover 9 state classes: closed, containing, empty, filled, folded, open, plugged, unfolded, unplugged. Following the OaSC release, four of these, namely empty, open, folded, and filled, are held out as *unseen* and never appear during training; the remaining five (closed, containing, plugged, unfolded, unplugged) are the *seen* classes. Test images span all nine classes and are remapped through the released folder-to-global-label mapping so that predictions and ground truth share a single 9-way label space. All images are resized to 224×224 and normalized with ImageNet statistics before being passed to the visual backbone.

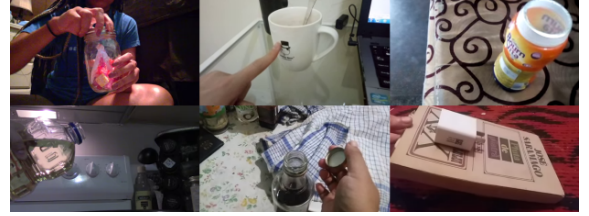


Fig. 4. A small sample of the OSDD dataset.

b) *MIT-States*.: MIT-States [3] is the classical compositional-state benchmark, containing state–object pairs such as *ripe tomato*, *sliced apple*, and *open door*. We follow the OaSC MIT-States configuration exactly, including the released seen/unseen state split. The object label is again discarded, so the model predicts only the state. MIT-States serves as the harder cross-dataset transfer setting in our study: **no hyperparameter, including the LRP residual weight α , is re-tuned on it.** All hyperparameters are frozen at their OSDD-selected values, which turns the MIT-States numbers into a genuine generalization test rather than a joint fit.

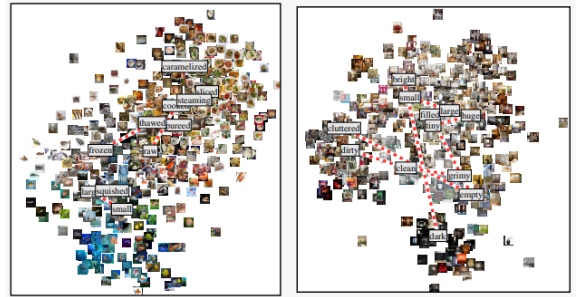


Fig. 5. A small sample of the MIT-States dataset.

All results are reported on the official OaSC test splits released with the paper.

B. Implementation Details

a) *Visual backbone*.: We use the ResNet-101 checkpoint released by the OaSC authors, which was fine-tuned on the OSDD training images with the Tr-GCN-generated state

vectors held fixed as the classifier weights. The backbone produces a 2048-dimensional pooled feature per image through global average pooling; we additionally retain the pre-GAP $7 \times 7 \times 2048$ layer-4 spatial map, which is required by the LRP module. ResNet-101 uses bottleneck residual blocks that make identity mappings easy to learn, allowing 101 layers at 7.6 GFLOP; its layer-4 feature map has an effective receptive field spanning most of the input, which makes even a coarse 7×7 grid meaningful for spatial reasoning about state.

b) Knowledge graph and Tr-GCN.: We use the released `conceptnet-wordnet-hop1-thresh-10` graph, which combines ConceptNet common-sense edges with WordNet lexicographic edges under a relation-weight threshold. To avoid the deprecated AllenNLP 0.9 dependency of the original OaSC code, we reimplement the two-layer Transformer GCN in pure PyTorch, matching AllenNLP’s `LayerNorm` (which uses the biased standard deviation) and its single-head `MultiHeadSelfAttention` layout. Layer 1 maps 300-D GloVe node features to 2048-D using a top-100 random-walk neighbor sampler, LeakyReLU (slope 0.2), and ℓ_2 normalization; Layer 2 maps 2048-D to 2049-D (2048 classifier weights + 1 bias) using a top-50 sampler and ℓ_2 normalization. We validate the reimplementation by regenerating the OSDD state classifiers from the *original* graph and comparing them against the released embedding file: **mean cosine similarity is ≥ 0.9995 across all nine states**, with maximum per-class deviation within the numerical tolerance of the released checkpoint. Because every graph-refinement experiment feeds modified graphs through *this same* Tr-GCN, any change in the classifier vectors is attributable to the change in graph and to nothing else.

c) CLIP text embeddings.: We use CLIP ViT-B/32 [12] (`openai/clip-vit-base-patch32`) as the sole external visual-language prior. Every KG node is embedded with the prompt "a visual concept related to <node>"; every target state is embedded with "a photo of an object that is <state>". All embeddings are ℓ_2 -normalized so that dot products correspond to cosine similarities. The CLIP model is frozen: no fine-tuning is performed at any stage, and the visual tower of CLIP is never used; only the text tower contributes to our pipeline.

d) Graph reranking.: For a target state s with unique neighbors n , we compute (i) a min-max normalized random-walk relevance $\hat{r}_{s,n}$ from the original graph and (ii) a normalized CLIP semantic relevance $\hat{c}_{s,n}$. We compare three fusion strategies for reranking the Tr-GCN neighbor list:

- **Fixed CLIP**: the normalized relevance scores are combined using constant weights,

$$q_{s,n} = 0.20, \hat{r}_{s,n} + 0.80, \hat{c}_{s,n}. \quad (27)$$

- **Adaptive CLIP**: the CLIP contribution is controlled by the per-state weight $\lambda_s \in [0.40, 0.90]$, which is derived from the ambiguity score

$$A_s = H_s D_s, \quad (28)$$

where H_s is the normalized entropy of the random-walk distribution over the n_s unique neighbors and $D_s = \log(1 + n_s) / \log(1 + N_{\max})$ is the neighborhood-size factor. The resulting reranking score is

$$q_{s,n}^{\text{adapt}} = (1 - \lambda_s) \hat{r}_{s,n} + \lambda_s \hat{c}_{s,n}. \quad (29)$$

- **OCGG (ours)**: the absolute CLIP similarity is replaced by the opposition-aware contrastive margin

$$m_{s,n} = \cos(\mathbf{e}_s, \mathbf{e}_n) - \cos(\mathbf{e}_{\bar{s}}, \mathbf{e}_n), \quad (30)$$

where $\bar{s}$ denotes the semantic opposite of state s . We use the four natural opposite pairs (open/closed, empty/filled, folded/unfolded, and plugged/unplugged). For the singleton state containing, which has no natural opposite, we retain the absolute similarity $c_{s,n}$. After neighborhood-wise normalization, OCGG uses

$$q_{s,n}^{\text{OCGG}} = (1 - \lambda_s) \hat{r}_{s,n} + \lambda_s \hat{m}_{s,n}. \quad (31)$$

Because OCGG uses the same adaptive weights as Adaptive CLIP, their performance difference isolates the contribution of the contrastive signal.

Critically, the reranked neighbor list is *not* truncated during reranking. The original Tr-GCN samplers, top-100 in Layer 1 and top-50 in Layer 2, still select the aggregation neighborhoods. Reranking therefore provides a soft reordering of the concepts entering aggregation rather than hard pruning, preserving compatibility with the pretrained Tr-GCN architecture.

e) Local Region Pooling (LRP).: For each image x , we retain the final spatial feature map $\mathbf{F}(x) \in \mathbb{R}^{2048 \times 7 \times 7}$ produced by ResNet-101 layer 4. For every candidate state s , we apply its Tr-GCN-generated classifier weight $\mathbf{w}_s$ at each spatial location to obtain the state-specific evidence map

$$M_s(h, w) = \mathbf{w}_s^\top \mathbf{F}[:, h, w(x)]. \quad (32)$$

In the implementation, this contraction is computed efficiently for all images and states using a batched `einsum` operation. This is the same weight-map contraction used in Class Activation Mapping [11]: the classifier weights that produce the global logit are reused to localize state-specific visual evidence. The classifier bias is excluded from $M_s(h, w)$ because it contains no spatial information and is already included in the global logit.

We then apply 3×3 average pooling with stride 1 to obtain a 5×5 grid of coherent region scores for each state. The three highest-scoring regions are selected and averaged to produce the local score L_s . The spatial mean of the evidence map is

$$G_s^{\text{vis}} = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W M_s(h, w). \quad (33)$$

Because the global logit z_s^{global} already contains this mean visual evidence, we subtract G_s^{vis} from L_s to isolate the localization residual. The final LRP-enhanced logit is

$$z_s^{\text{LRP}} = z_s^{\text{global}} + \alpha(L_s - G_s^{\text{vis}}). \quad (34)$$

As a sanity check, global averaging of $\mathbf{F}(x)$ reproduces the 2048-dimensional feature used by the baseline classifier. The mean cosine similarity between the original GAP output and the spatial mean of $\mathbf{F}(x)$ is 1.0, with a mean absolute difference of 0.0. This confirms that LRP operates on the pre-GAP representation underlying the baseline prediction rather than on a different visual feature.

We report $\alpha = 0.20$ as the operating configuration identified in the post-hoc OSDD sensitivity analysis over $0.00, 0.025, \dots, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00$ (Fig. 6). The same $\alpha = 0.20$ configuration is then frozen and transferred to MIT-States without retuning, making the MIT-States evaluation a cross-dataset generalization test. The per- α results and the interaction between LRP and OCGG are discussed in Section V-D.

f) *Compute and determinism.*: All experiments run on a single GPU with inference batch size 128. Because the CNN, Tr-GCN, and CLIP are all frozen, the pipeline is fully deterministic given inputs: OSDD and MIT-States test features are each cached once and reused across every semantic-only variant, and LRP components (global logit z_s^{global} and localization gain $L_s - G_s^{\text{vis}}$) are cached per method so that the α sensitivity sweep does not require additional CNN forward passes.

C. Evaluation Metrics

We follow the generalized zero-shot state-classification protocol of the OaSC paper, which itself adapts the bias-sweep evaluation of Purushwalkam et al. [5]. For each image the model produces a score vector over all state classes; a scalar bias γ is then added to the scores of the unseen classes before the argmax:

$$\hat{y}(x) = \arg \max_s [\text{score}(x, s) + \gamma \cdot \mathbb{I}[s \in \mathcal{S}^U]]. \quad (35)$$

Sweeping γ over $[-5.0, 15.0]$ in steps of 0.5 traces out an operating curve between the seen-favoring regime (large negative γ) and the unseen-favoring regime (large positive γ). From this curve we report four metrics, all as percentages and all matched to the definitions used in Table 1 of Goudis et al. [2] so that our numbers are directly comparable to the OaSC paper.

a) *S: Best Seen Accuracy.*: The maximum accuracy on seen-state test images over the γ sweep. This is realized at strongly negative γ , where the model prioritizes seen classes. It measures how well the classifier separates seen states from each other when the seen–unseen trade-off is resolved in favor of seen classes.

b) *U: Best Unseen Accuracy.*: The corresponding maximum on unseen-state test images, realized at large positive γ . It measures how well the graph-generated templates for held-out states discriminate them from each other and from seen classes when the calibration bias is turned to their advantage.

c) *HM: Best Harmonic Mean.*: The single best trade-off point on the operating curve,

$$\text{HM} = \max_{\gamma} \frac{2 \cdot S(\gamma) \cdot U(\gamma)}{S(\gamma) + U(\gamma)}. \quad (36)$$

HM is our headline metric because it penalizes any method that achieves high accuracy on one side of the seen–unseen partition only by collapsing the other. Note that the γ that maximizes HM is generally different from the γ that maximizes S or U in isolation.

d) *A: AUC.*: The area under the seen-vs-unseen accuracy curve, computed by the trapezoidal rule over the γ grid. AUC characterizes the *entire* operating curve rather than a single point and is therefore more robust than HM to the choice of γ . Two methods with similar HM can have visibly different AUC if one has a wider region of good trade-offs and the other only a narrow sweet spot.

We report all four numbers side-by-side because they answer different questions: S and U diagnose which side of the partition a method is strong on, HM summarizes the best achievable balance, and A summarizes robustness across operating points.

D. Compared Variants

We report six variants that share the OaSC backbone, released fine-tuned checkpoint, Tr-GCN weights, and evaluation protocol, and differ only in their graph-refinement and/or visual-pooling components. This holds all uncontrolled sources of variation fixed and makes every reported delta attributable to a single design change.

- 1) **Original OaSC.** Released state classifiers used as-is; global GAP feature only. This is our reproduction baseline and matches the numbers reported by Goudis et al. [2] within numerical precision.
- 2) **Fixed CLIP KG.** Graph reranked with the constant ($w_{\text{RW}} = 0.20$, $w_{\text{CLIP}} = 0.80$) fusion; global GAP feature only.
- 3) **Adaptive CLIP KG.** Graph reranked with the ambiguity-adaptive per-state CLIP weight λ_s ; global GAP feature only.
- 4) **OCGG (ours).** Graph reranked with the opposition-aware contrastive score $m_{s,n} = \cos(\mathbf{e}_s, \mathbf{e}_n) - \cos(\mathbf{e}_{\bar{s}}, \mathbf{e}_n)$ fused under the same adaptive weights as (3); global GAP feature only.
- 5) **OaSC + LRP ($\alpha = 0.20$).** Original OaSC classifiers with the LRP residual added at inference. Isolates the effect of the visual pathway on the un-modified baseline.
- 6) **OCGG + LRP ($\alpha = 0.20$).** The full proposed method: OCGG-refined classifiers with the LRP residual.

Variants (1)–(4) hold the visual pathway fixed and isolate the effect of graph refinement. Comparing (1) vs. (5) and (4) vs. (6) isolates the additive contribution of LRP once the graph pathway is fixed. The gap between (5) and (6) at fixed α measures how much of the improvement remains attributable to opposition-aware graph reranking once the visual localization component is already in place. On MIT-States the same six variants are reported with α transferred from OSDD unchanged, so the MIT-States column is a cross-dataset generalization test rather than a joint fit.

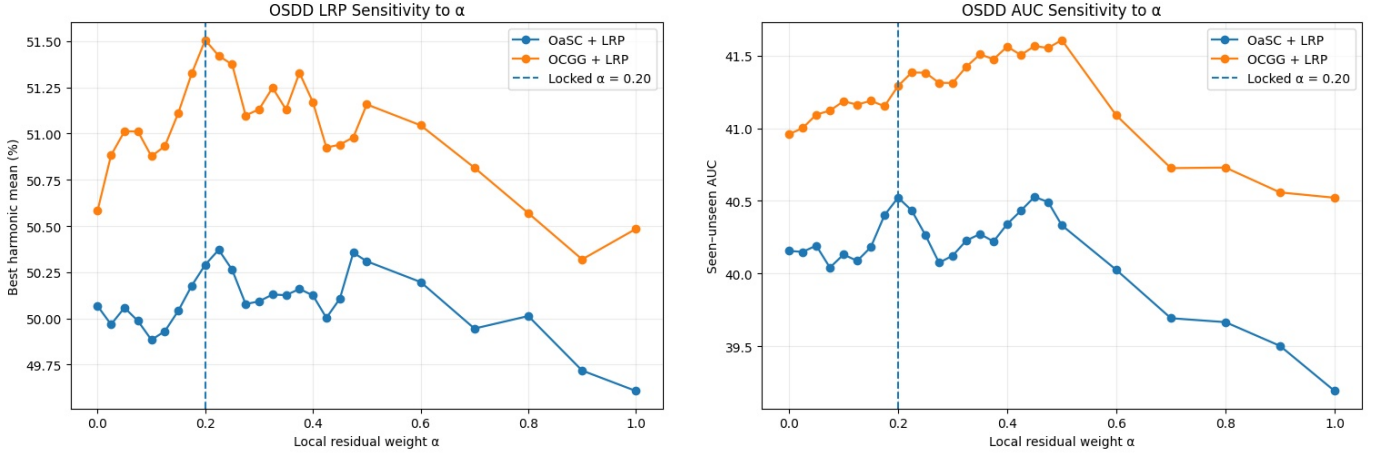


Fig. 6. Post-hoc OSDD test-set sensitivity of LRP to the residual weight α . The vertical line marks the reported $\alpha = 0.20$ configuration. Because this analysis uses the test split, it is reported as an exploratory sensitivity analysis rather than validation-based hyperparameter selection.

TABLE I
COMPATIBILITY OF THE RECONSTRUCTED Tr-GCN WITH THE RELEASED OASC STATE VECTORS.

Measure	Value
Mean cosine similarity	0.999928
Minimum cosine similarity	0.999476
Mean absolute error	0.00013837
Maximum absolute error	0.00334771

V. RESULTS AND ANALYSIS

All variants in this section are evaluated with the same dataset splits, state-label space, calibrated-stacking procedure, and released ResNet-101 visual backbone. Unless otherwise stated, only the semantic classifier vectors or the Local Region Pooling (LRP) score correction are changed. This keeps the comparisons compatible and lets the ablations isolate the effect of each proposed component.

A. Reproducing OaSC / Tr-GCN Verification

Before modifying the graph, we first verified that our pure-PyTorch reconstruction of the released Transformer Graph Convolutional Network (Tr-GCN) reproduces the original OaSC classifier vectors [2]. On the nine OSDD states, the reconstructed vectors have a mean cosine similarity of 0.999928 with the released vectors, while the minimum state-wise cosine similarity is 0.999476. The mean and maximum absolute errors are 1.3837×10^{-4} and 3.34771×10^{-3} , respectively. Table I summarizes this compatibility check. The close agreement is important because it makes later changes attributable to the graph reranking rather than to a different implementation of Tr-GCN.

Using the released OaSC classifier and checkpoint, the reproduced OSDD baseline obtains a best harmonic mean (HM) of 50.07%, with 44.68% seen accuracy and 56.93% unseen accuracy at the HM-optimal calibration value $\gamma = 2.5$. Its maximum unseen accuracy is 64.82% and its seen-unseen

AUC is 40.16. These reproduced values are used as the reference point for all OSDD improvements below.

B. Semantic Graph Ablation

We next isolate the effect of semantic graph refinement and Local Region Pooling (LRP). Table II compares the original OaSC graph with fixed CLIP fusion, ambiguity-adaptive CLIP fusion, the proposed Opposition-Aware Contrastive Graph Gating (OCGG), and the corresponding LRP variants. The same Tr-GCN and visual features are used for every row.

On OSDD, all three CLIP-guided variants improve the baseline HM, and OCGG gives the highest semantic-only HM (50.58%), maximum unseen accuracy (66.41%), and AUC (40.96). The gain over the original graph is modest but consistent across these three metrics. On MIT-States, however, fixed CLIP fusion is the strongest semantic-only variant, reaching 54.58% HM and 45.57 AUC, while OCGG reaches 53.64% HM and 43.47 AUC. Therefore, the semantic ablation does not support the claim that opposition-aware reranking is uniformly better by itself. Instead, OCGG is most useful in the final system when combined with the spatial correction introduced by LRP.

C. OCGG Mechanism Analysis

The mechanism analysis explains why absolute semantic similarity can be insufficient for opposite states. For the target state *open*, the concept *closed* is highly related in an ordinary semantic space even though it provides evidence for the opposite visual condition. Under the adaptive CLIP graph, *closed* is ranked first in the *open* neighborhood. After opposition-aware scoring, it falls to rank 189. Similarly, *close* moves from rank 3 to rank 186. In the CLIP comparison, *closed* has similarity 0.8698 to the target prompt for *open* but 0.8844 to the prompt for *closed*, producing a negative contrastive margin of -0.0147 .

Fig. 7 summarizes this behavior. Across the *open* neighborhood, the contrastive margin and OCGG rank change have a

TABLE II

SEMANTIC GRAPH AND LRP ABLATION. ALL VALUES ARE PERCENTAGES EXCEPT AUC, WHICH IS REPORTED ON THE SAME 0–100 SCALE USED IN THE NOTEBOOK.

Dataset	Semantic graph	Max Seen	Max Unseen	Best HM	AUC
OSDD	Original OaSC	84.46	64.82	50.07	40.16
OSDD	Fixed CLIP KG	84.67	66.16	50.24	40.76
OSDD	Adaptive CLIP KG	84.57	65.74	50.56	40.63
OSDD	OCGG	84.36	66.41	50.58	40.96
OSDD	OaSC + LRP	84.36	65.66	50.29	40.52
OSDD	OCGG + LRP	84.10	66.75	51.51	41.29
MIT-States	Original OaSC	85.71	69.87	51.37	40.99
MIT-States	Fixed CLIP KG	85.71	75.00	54.58	45.57
MIT-States	Adaptive CLIP KG	85.71	75.00	54.41	45.50
MIT-States	OCGG	83.33	75.00	53.64	43.47
MIT-States	OaSC + LRP	88.10	69.87	50.68	39.87
MIT-States	OCGG + LRP	88.10	75.64	57.23	46.33

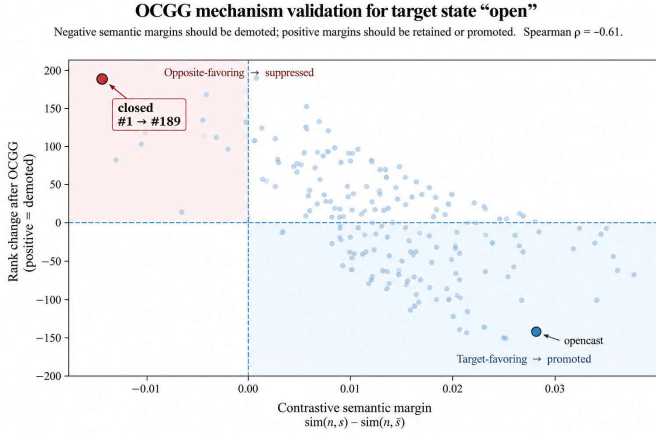


Fig. 7. OCGG mechanism analysis for the state *open*. The contrastive margin is associated with the direction of rank change; using the notebook’s rank-drop convention, the Spearman correlation is -0.6084 .

Spearman correlation of -0.6084 under the notebook’s rank-drop convention: concepts that favor the target tend to move toward better ranks, while concepts that favor the opposite tend to be suppressed. At the same time, the promoted list still contains some lexically related but visually questionable concepts. OCGG therefore addresses a specific source of semantic confusion—state opposition—but it does not remove every noisy concept in a general-purpose knowledge graph.

D. Local Region Pooling and α Sensitivity

The sensitivity curves in Fig. ?? show different behavior for the original OaSC classifier and OCGG. For OCGG, the highest observed HM is 51.51% at $\alpha = 0.20$; the highest observed AUC occurs at $\alpha = 0.50$ with 41.61, compared with 41.29 at $\alpha = 0.20$. For the original OaSC classifier, the highest observed HM is 50.37% at $\alpha = 0.225$. Because these values were measured on the OSDD test set, we report the sweep only as sensitivity analysis. The final $\alpha = 0.20$ setting is then frozen and transferred to MIT-States without retuning.

The interaction in Fig. 9 is important. On OSDD, adding LRP to the original OaSC classifier changes HM from 50.07% to 50.29% (+0.22 percentage points), while adding it to

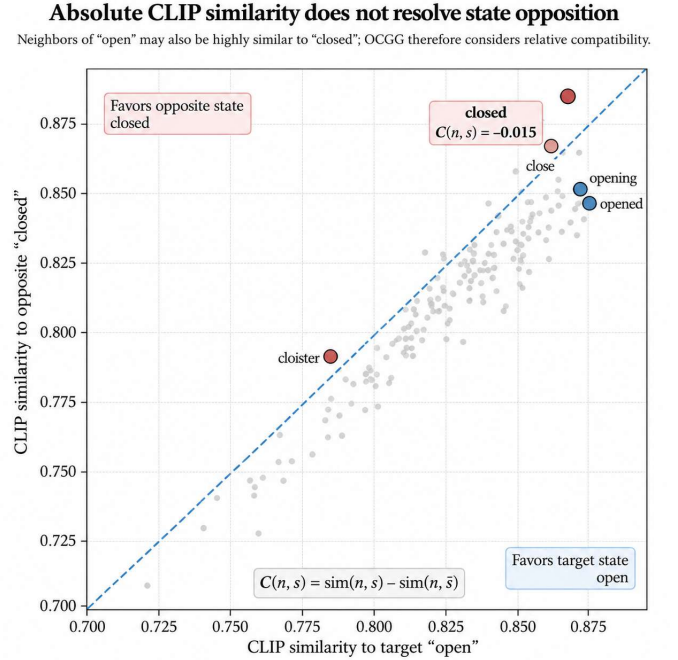


Fig. 8. Absolute CLIP similarity for neighbors of *open*. The explicit opposite *closed* remains highly similar to *open*; the contrastive margin reveals that it is even more compatible with the opposite state.

OCGG changes HM from 50.58% to 51.51% (+0.92 points). On MIT-States, LRP alone changes the OaSC HM from 51.37% to 50.68%, but applying the same frozen LRP configuration to OCGG increases HM from 53.64% to 57.23%. Thus, LRP is not a universally beneficial post-processing step; its effect depends on the classifier whose spatial evidence it refines.

E. OSDD Results

Table III compares the final OCGG+LRP model with prior methods reported by Goudis *et al.* [2], as well as our reproduced OaSC baseline. On OSDD, OCGG+LRP achieves the highest HM of 51.51%, improving the reproduced OaSC baseline by +1.44 percentage points. Maximum unseen-state accuracy increases from 64.82% to 66.75%, while AUC in-

LRP becomes more effective after semantic refinement

The same local-region module is applied to both classifiers; the larger gain with OCGG indicates a semantic-spatial interaction.

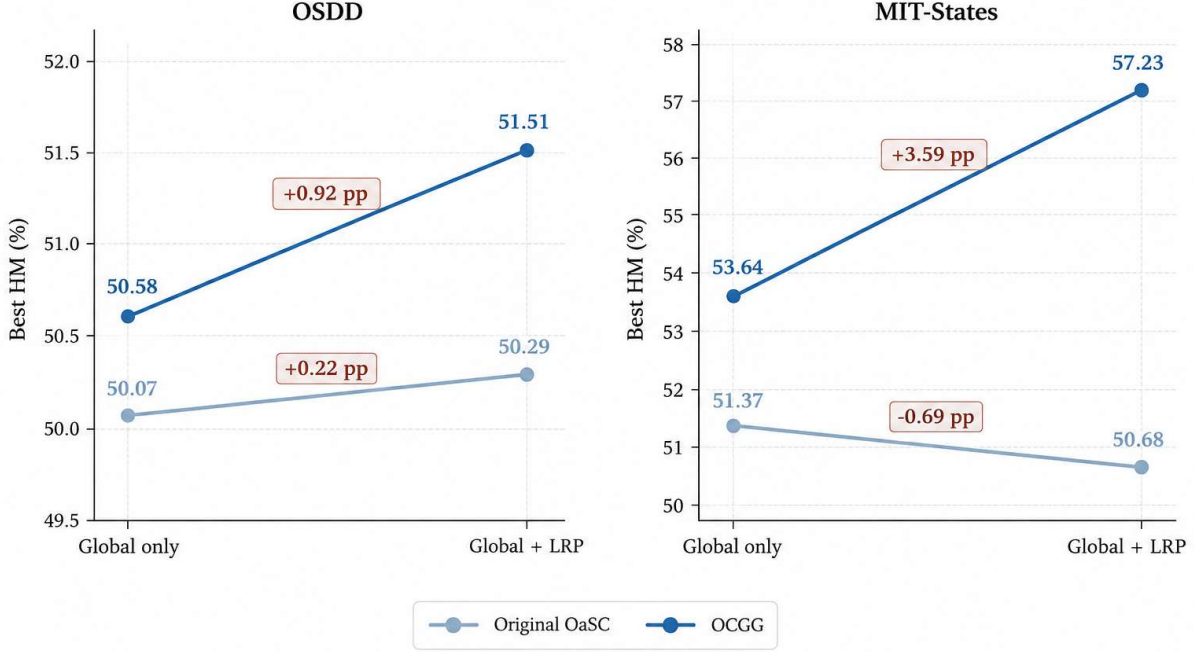


Fig. 9. Interaction between semantic refinement and LRP. LRP gives only a small OSDD gain on the original OaSC classifier and slightly reduces its MIT-States HM, whereas the same $\alpha = 0.20$ correction provides a larger improvement when applied after OCGG.

creases from 40.16 to 41.29. On MIT-States, OCGG+LRP improves HM from 51.37% to 57.23%, with maximum unseen accuracy increasing from 69.87% to 75.64% and AUC from 40.99 to 46.33. These results show that the proposed semantic and spatial refinements improve the reproduced OaSC baseline on both datasets, with a substantially larger gain on MIT-States.

The improvement is not simply an increase in the unseen accuracy measured at the HM-optimal operating point. At that point, unseen accuracy changes from 56.93% to 56.26%, while seen accuracy increases from 44.68% to 47.50%. The resulting balance raises HM. At the same time, the maximum unseen accuracy over the calibration sweep rises by 1.93 points and the AUC rises by 1.14, indicating that the final method improves the broader seen–unseen trade-off rather than only one operating point. OCGG and LRP individually yield smaller gains than their combination, which supports using the semantic and local visual refinements together in the final OSDD system.

F. MIT-States Transfer

MIT-States is used as a cross-dataset transfer test for the final visual configuration. The 3×3 region size, top-3 pooling, and $\alpha = 0.20$ residual weight are copied directly from OSDD. Table III shows that OCGG+LRP reaches 57.23% HM, compared with 51.37% for the original OaSC baseline.

Relative to the original OaSC baseline, the final method improves HM by 5.85 points, maximum unseen accuracy by 5.77 points, and AUC by 5.33. Relative to OCGG alone, LRP improves HM by 3.59 points and raises unseen accuracy at the HM-optimal point from 44.87% to 53.21%, while seen accuracy decreases from 66.67% to 61.90%. In other words, LRP substantially improves the balance of the OCGG classifier on MIT-States. The fact that OaSC+LRP alone decreases HM and AUC further emphasizes that the local correction is most effective in conjunction with the refined semantic classifier.

The calibration curves in Fig. 10 provide a fuller view of the generalized zero-shot behavior. On OSDD the four curves remain close, matching the relatively small numerical gains in Table II. On MIT-States the separation is larger, and OCGG+LRP maintains a more favorable seen–unseen trade-off over a broader part of the calibration sweep, consistent with its higher AUC.

G. Qualitative Analysis

The qualitative LRP visualizations show both the behavior that improves predictions and the cases in which local evidence is misleading. The heatmaps visualize positive classifier evidence for the ground-truth state, while the black boxes show the top-scoring 3×3 regions used by the local pooling rule. These visualizations are diagnostic views of the fixed classifier scores; they are not a separately learned attention module.

TABLE III

AGGREGATE RESULTS FOR THE OBJECT AGNOSTIC SETTING ON OSDD AND MIT-STATES. RESULTS FOR PRIOR METHODS ARE TAKEN FROM GOUIDIS *et al.* [2]. S AND U DENOTE THE BEST ACCURACIES ON SEEN AND UNSEEN CLASSES, RESPECTIVELY; HM DENOTES THE BEST HARMONIC MEAN, AND A DENOTES THE AREA UNDER THE SEEN-UNSEEN ACCURACY CURVE. THE GAIN ROW REPORTS THE IMPROVEMENT OF OCGG+LRP OVER OUR REPRODUCED OASC BASELINE.

2*Method	OSDD				MIT-States			
	S	U	HM	A	S	U	HM	A
AoP [15]	43.2	26.1	20.7	7.4	100.0	11.6	13.2	7.0
LE+ [16]	30.5	31.9	14.0	4.3	100.0	20.5	19.4	10.0
TMN [5]	83.1	66.5	38.5	27.5	100.0	17.8	17.7	11.0
SymNet [17]	83.8	37.3	33.5	19.8	94.1	21.4	23.2	13.2
CompCos [18]	86.5	43.7	26.9	15.9	100.0	52.2	36.4	25.8
KG-SP [19]	80.0	39.8	26.7	12.4	100.0	7.1	9.0	4.0
SCEN [20]	77.8	41.5	35.2	22.5	100.0	22.03	20.6	12.6
IVR [21]	85.8	37.8	35.1	22.1	100.0	11.3	14.1	5.4
OADiS [22]	72.7	55.5	23.1	13.0	94.1	30.0	23.3	12.5
CANET [23]	85.6	36.4	20.2	12.1	100.0	16.9	23.1	11.9
ADE [24]	91.4	67.1	40.5	30.3	100.0	24.9	22.6	12.6
<i>OaSC (reproduced)</i>	84.46	64.82	50.07	40.16	85.71	69.87	51.37	40.99
OCGG + LRP (ours)	84.10	66.75	51.51	41.29	88.10	75.64	57.23	46.33
Δ (gain vs. reproduced OaSC)	–	+1.93	+1.44	+1.14	+2.39	+5.77	+5.85	+5.33

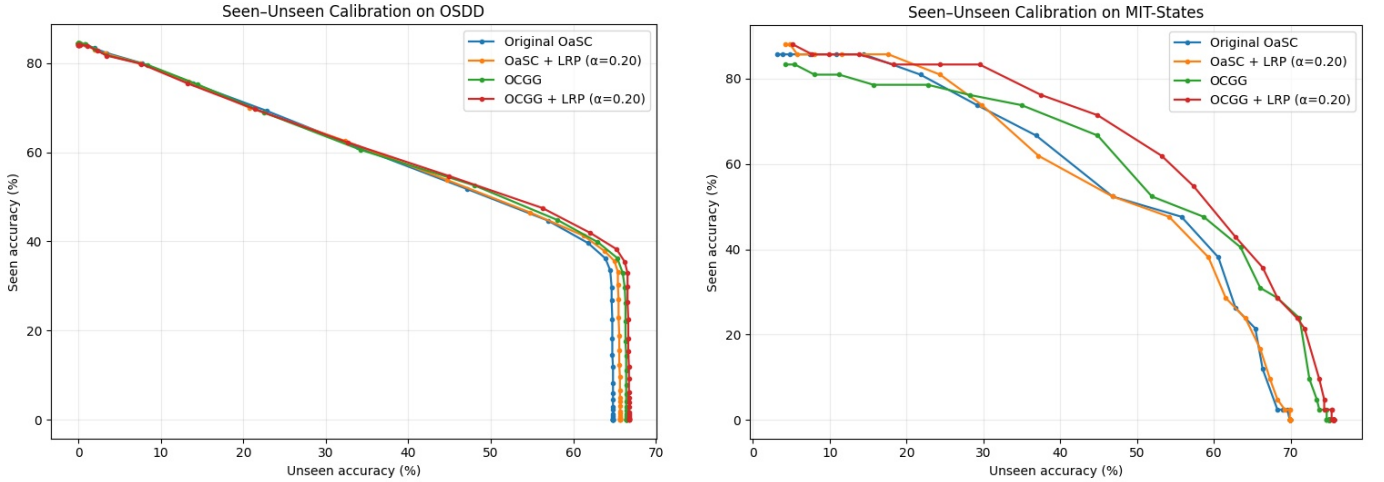


Fig. 10. Seen-unseen calibration curves for OSDD (left) and MIT-States (right). The curves show the complete calibrated-stacking trade-off rather than only the best-HM operating point.

The OSDD analysis contains 78 examples that are misclassified by OCGG but corrected by OCGG+LRP. Three representative rescued cases are shown in Fig. 13: *unplugged* is corrected from *open*, *plugged* from *empty*, and *closed* from *open*. Their ground-truth-vs.-competitor margin gains are 1.320, 0.882, and 0.612, respectively.

Table IV summarizes the per-state effect of LRP on top of OCGG. The effect is heterogeneous. On OSDD, LRP improves several seen states, including *unfolded* (+4.96) and *closed* (+4.83) points, but reduces accuracy for some unseen states, most notably *filled* (−4.08) points. On MIT-States, the largest gains occur for unseen *closed*, *filled*, and *folded*, whereas the seen state *empty* decreases by 12.00 points. These state-level changes are consistent with the success and failure cases in Figs. 11 and 12: localized evidence can be highly informative for some states but can also emphasize visually salient regions that support a competing state.

Overall, the quantitative and qualitative results support a restrained conclusion. OCGG improves the semantic graph on OSDD and explicitly suppresses a major source of confusion between opposing states, but it is not the strongest semantic-only variant on every dataset. LRP is likewise not uniformly beneficial when attached to the original OaSC classifier. The most consistent result is obtained when the two modifications are used together: OCGG+LRP gives the best HM among the evaluated final variants on both OSDD and MIT-States, with the larger gain appearing in the cross-dataset MIT-States experiment.

VI. DISCUSSION AND LIMITATIONS

A. Semantic and Visual Refinements Are Complementary

The most consistent quantitative finding of this study is not that either component wins in isolation, but that the two components are complementary. On OSDD, semantic-only

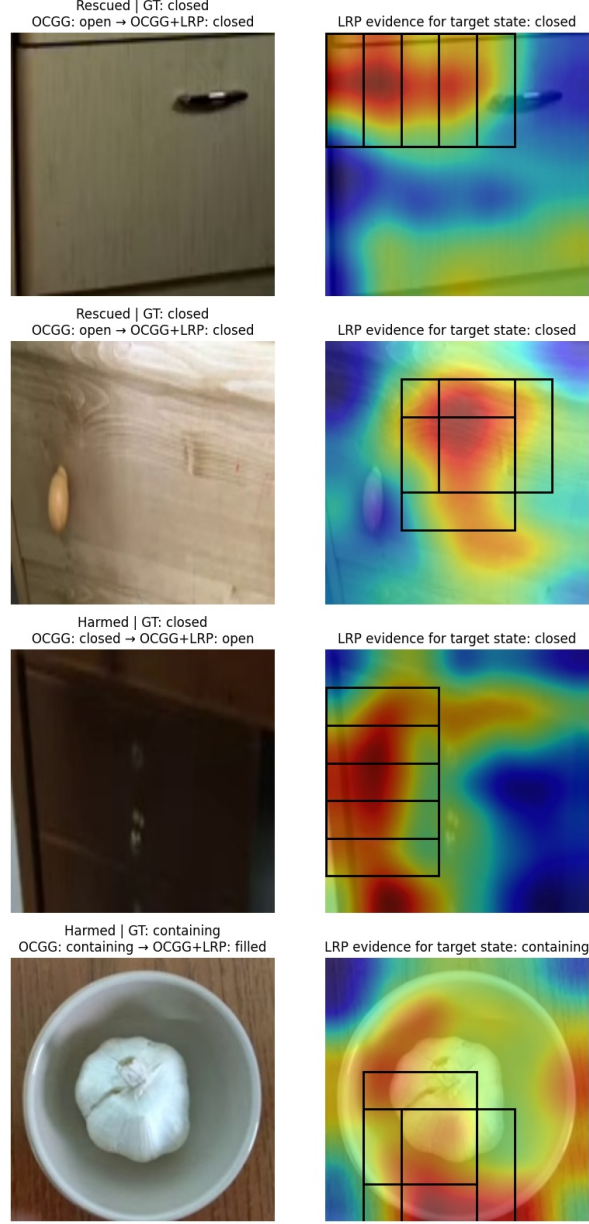


Fig. 11. Representative OSDD LRP transitions. The first two examples are rescued from *open* to the correct state *closed*. The lower examples are failure cases where LRP changes an initially correct prediction to *open* or *filled*.

OCGG improves HM by 0.51 points over the reproduced OaSC baseline, and LRP alone improves HM by 0.22 points; used together, the gain is +1.44 points. On MIT-States, semantic-only OCGG improves HM by 2.27 points and LRP alone actually decreases HM by 0.69 points, but the combination improves HM by 5.85 points. The joint improvement is therefore larger than the sum of the individual improvements on both datasets, which suggests that opposition-aware graph reranking and localized visual pooling address related but distinct sources of error.

A plausible mechanism is that OCGG produces classifier

vectors whose spatial responses are more state-discriminative. In the baseline OaSC classifier, the last-layer weights for a state such as *open* still carry evidence associated with lexically related but visually opposite concepts such as *closed*. When these weights are contracted with the 7×7 layer-4 feature map, the highest-scoring local regions can correspond to visual evidence that supports the opposite state, and LRP’s top-3 region pooling then amplifies this misleading evidence rather than correcting it. Once OCGG has suppressed opposite-supporting concepts from the neighborhood that the Tr-GCN aggregates over, the resulting classifier’s spatial response is

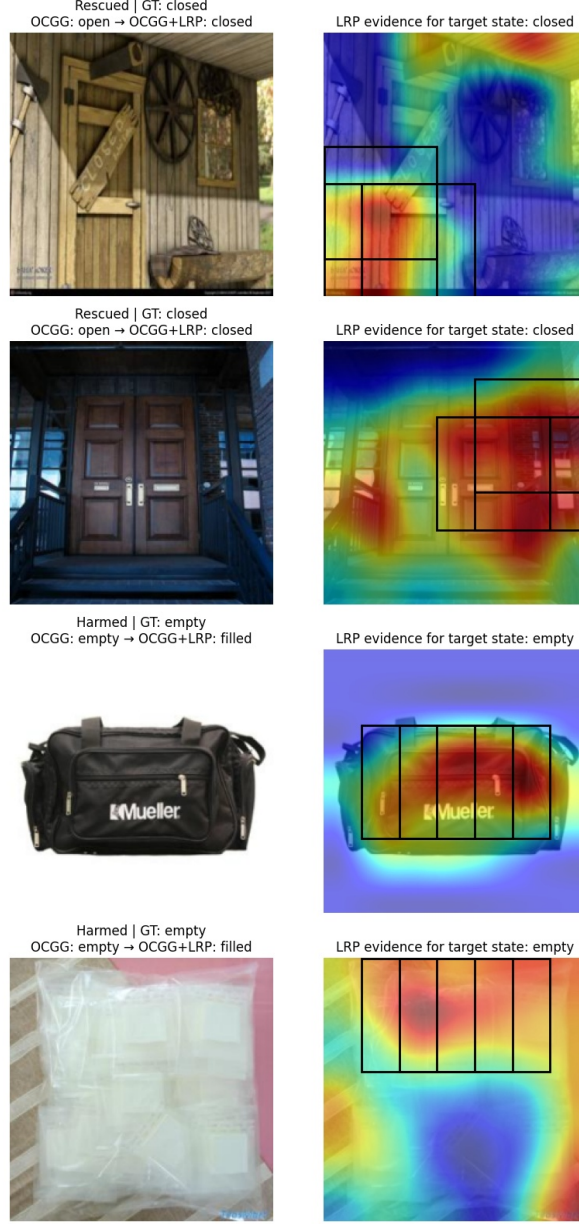


Fig. 12. Representative MIT-States LRP transitions. LRP correctly changes two *open* predictions to *closed*, but it can also overemphasize local evidence and change correctly classified *empty* examples to *filled*.

more state-specific, so pooling over the strongest local regions moves the prediction in the correct direction.

The strongest evidence for this interaction hypothesis is the MIT-States rows of Table II. LRP applied to the original OaSC classifier reduces HM by 0.69 points and AUC by 1.12, while applied to the OCGG classifier it increases HM by 3.59 points and AUC by 2.86. The two visual pipelines are otherwise identical, so the gap must originate in the classifier’s spatial response.

B. Why the Gain is Larger on MIT-States

The absolute improvement on MIT-States (+5.85 HM points) is substantially larger than on OSDD (+1.44 HM points). Two properties of MIT-States are consistent with this. First, its state vocabulary contains a larger fraction of concepts with natural opposites, including *ripe/unripe*, *cooked/raw*, *sliced/whole*, *wet/dry*, and *full/empty*. OCGG can therefore act on a broader set of states through its opposite pairs, and the opposition-density hypothesis is a testable prediction of our method: its benefit should scale with the fraction of test states that have lexically opposed distractors. Second, MIT-States

LRP-rescued predictions: localized state evidence corrects OCGG errors

Examples are automatically selected where OCGG is wrong but OCGG+LRP is correct; different target states are preferred.

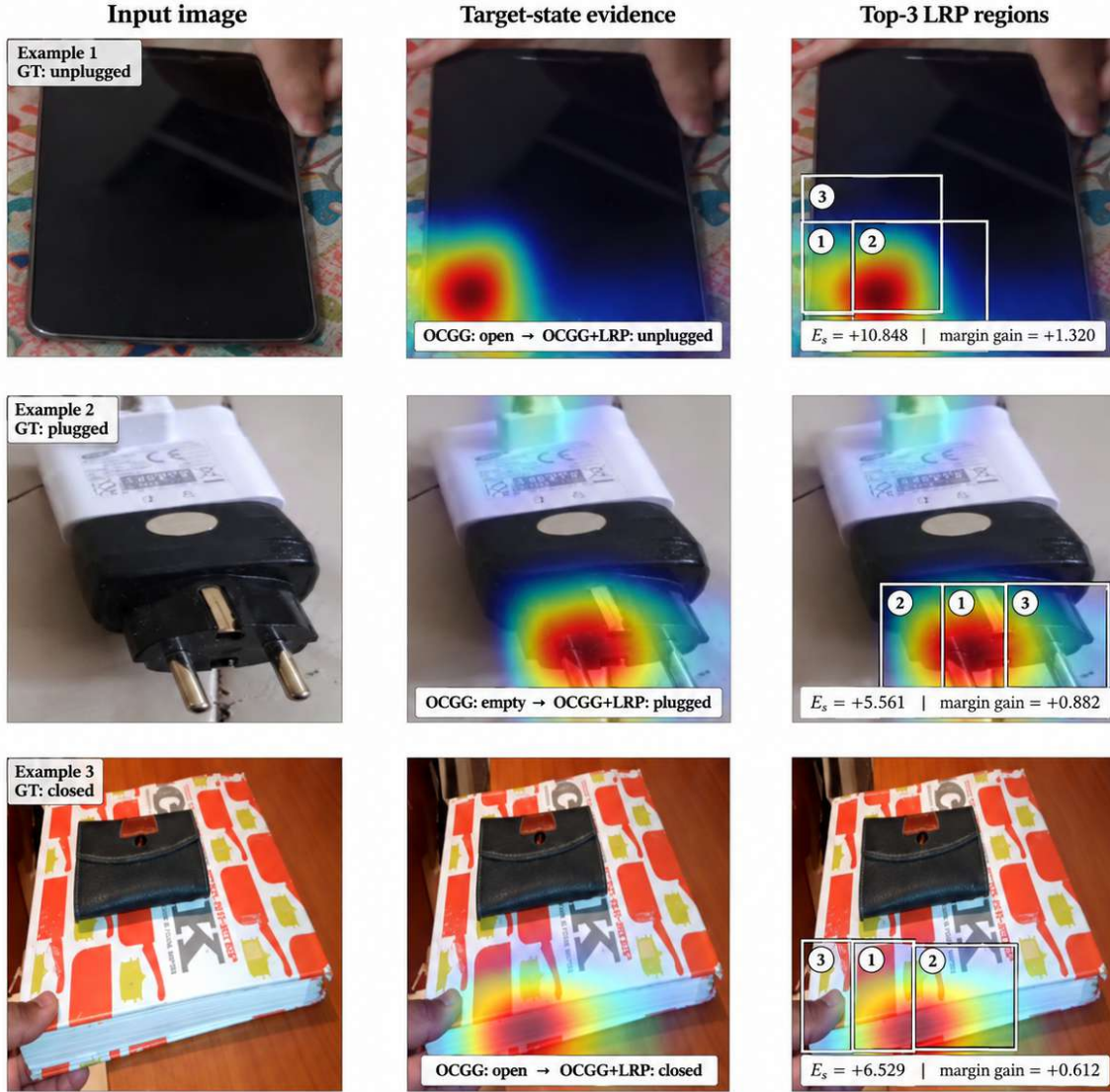


Fig. 13. Three OSDD examples rescued by LRP after OCGG. Each row shows the original image, target-state evidence, and the top local regions used by LRP.

spans a much wider range of object categories, so object-agnostic reasoning is more heavily pressed and object identity offers less of a shortcut. This favors methods whose classifiers do not implicitly rely on the object appearance shared by seen states.

C. Per-State Heterogeneity

Table IV exposes a limitation that is not visible in the aggregate metrics. On OSDD, LRP on top of OCGG improves several seen states, notably *unfolded* (+4.96) and *closed* (+4.83), but reduces the unseen *filled* accuracy by 4.08 points. On MIT-States, unseen *closed*, *filled*, and *folded* improve

substantially, while the seen state *empty* decreases by 12.00 points. Two structural reasons help explain this. States defined by the absence of content, such as *empty*, are inherently non-local: their signature is the lack of a filled region, which is not well captured by top-3 region pooling that amplifies the strongest positive evidence. States defined by localized manipulations, such as *folded* or *plugged*, are better matched to LRP’s inductive bias. A method that adapts the residual weight α per state, or replaces top- K pooling with an absence-aware operator, is a natural direction for reducing the downside on absence-defined states.

Small class sizes also enlarge the variance of per-state

TABLE IV
PER-STATE ACCURACY BEFORE AND AFTER LRP ON TOP OF OCGG. Δ IS THE CHANGE IN PERCENTAGE POINTS.

Dataset	State	Count	Split	OCGG	OCGG+LRP	Δ
OSDD	unfolded	121	Seen	3.31	8.26	+4.96
OSDD	closed	559	Seen	10.73	15.56	+4.83
OSDD	unplugged	681	Seen	84.43	86.64	+2.20
OSDD	containing	297	Seen	21.55	23.23	+1.68
OSDD	plugged	260	Seen	60.00	59.62	-0.38
OSDD	empty	647	Unseen	61.82	60.59	-1.24
OSDD	open	207	Unseen	69.08	67.63	-1.45
OSDD	folded	141	Unseen	82.27	80.14	-2.13
OSDD	filled	196	Unseen	16.84	12.76	-4.08
MIT-States	closed	77	Unseen	11.69	22.08	+10.39
MIT-States	filled	42	Unseen	11.90	21.43	+9.52
MIT-States	folded	193	Unseen	65.28	72.54	+7.25
MIT-States	open	17	Seen	29.41	35.29	+5.88
MIT-States	empty	25	Seen	92.00	80.00	-12.00

numbers on MIT-States. The seen state *empty* contains only 25 test examples in our split, so a swing of three correct predictions moves per-state accuracy by 12 points. Aggregate metrics such as HM and AUC are less sensitive to this, which is one reason we report them alongside the per-state breakdown.

D. Limitations

a) Test-set hyperparameter sensitivity.: The residual weight $\alpha = 0.20$ is selected by a sensitivity sweep on the OSDD test set (Fig. ??). As noted in Section V-D, we report this only as a sensitivity analysis rather than validation-based hyperparameter selection, and we freeze α before evaluating on MIT-States so that the MIT-States column is a genuine transfer test. Nevertheless, an OSDD-only held-out validation split would be a stronger protocol and is straightforward to add given the released OSDD training folder.

b) Handcrafted opposition structure.: OCGG requires an explicit opposite state for each target that benefits from contrastive scoring. Our four pairs are handcrafted, and the state *containing* falls back to absolute similarity because it has no natural lexical opposite. Two extensions are natural. First, each state could be paired with a set of antonyms and the contrastive score computed against the most confusable member of that set. Second, opposition could be estimated directly from the CLIP or GloVe embedding space rather than being provided by hand, which would remove the reliance on a per-dataset opposition table.

c) Prompt sensitivity.: The CLIP similarity signal depends on a single handcrafted prompt for each state ("a photo of an object that is {state}") and a single prompt for each graph node. The pipeline is deterministic given these prompts, so classical seed variance is not the correct uncertainty measure, but prompt variance is. A prompt ensemble in the style of CLIP zero-shot classification [12] is a natural robustness check, and the small OSDD HM gain in particular (+1.44 points) is in a range where a single prompt choice could plausibly change conclusions.

d) Visual backbone.: We retain the OaSC ResNet-101 backbone and released checkpoint. LRP's quality depends on

the spatial feature map it operates on, and a stronger backbone with finer spatial structure, such as DINOv2 or CLIP-ViT, could improve the localization signal. This is not a free substitution in our framework because the Tr-GCN produces 2049-dimensional classifier vectors matched to the 2048-D ResNet feature dimension, so a different backbone would require either re-training the classifier-generation pipeline or introducing a projection layer.

e) Absence-defined states.: LRP's top- K region pooling is well matched to states whose evidence is localized (a plug, a fold, a lid), but poorly matched to states defined by the absence of a feature (*empty*) or by whole-object properties. This is visible in the per-state analysis and represents an intrinsic limitation of the pooling rule rather than a tuning issue.

f) Evaluation breadth.: Our evaluation covers OSDD [1] and MIT-States [3], which correspond to the two most commonly reported settings in the OaSC line of work. We do not report the C-GQA-States or VAW-States subsets used in the extended OaSC evaluation [2], so the transfer story remains a two-dataset argument. Broader evaluation is straightforward under the same protocol and is a natural next step.

g) Statistical characterization.: Because the CNN, Tr-GCN, and CLIP text encoder are all frozen and the pipeline is deterministic given inputs, we do not report seed-based confidence intervals. The uncertainties in our numbers therefore originate in prompt choice, dataset splits, and the discreteness of the γ grid used for the calibration sweep. Reporting these sensitivities explicitly, particularly for the smaller-margin OSDD results, would strengthen future versions of this work.

VII. CONCLUSION

We studied object-agnostic zero-shot state recognition and proposed two lightweight extensions of the OaSC framework. Opposition-Aware Contrastive Graph Gating (OCGG) reranks knowledge-graph neighborhoods so that CLIP-based concepts supporting a target state more strongly than its lexical opposite are favored before the Tr-GCN generates classifier vectors. Local Region Pooling (LRP) supplements the global visual score with residual evidence from the strongest local regions of the ResNet-101 feature map. Neither component retrains

the Tr-GCN or the visual backbone, so both operate as drop-in refinements of the released OaSC pipeline.

Across OSDD and MIT-States, the combination of OCGG and LRP improves the reproduced OaSC baseline in the generalized zero-shot setting, raising the harmonic mean from 50.07% to 51.51% on OSDD and from 51.37% to 57.23% on MIT-States, together with consistent gains in maximum unseen accuracy and AUC. The improvement is not carried by either component alone. On MIT-States, LRP applied to the unmodified OaSC classifier slightly reduces HM, while the same LRP configuration applied on top of OCGG produces a +3.59-point HM gain. This interaction, together with the larger overall gain on the opposition-denser MIT-States, suggests that opposition-aware semantic reasoning and localized visual evidence address related but distinct sources of error in object-agnostic state recognition.

Natural follow-up work, motivated by the limitations discussed in Section VI, includes replacing handcrafted opposition pairs with data-driven or embedding-based alternatives, extending the visual backbone to models with finer spatial structure, replacing top- K region pooling with an operator that also handles absence-defined states, and evaluating on broader benchmarks under an OSDD-only validation protocol. We believe the two-component recipe of opposition-aware graph gating and residual local pooling is a useful starting point for follow-up work on graph-guided object-agnostic state recognition.

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